

SavePredictedExams — SME

Economics

Higher level

Paper 3

SME Mark Scheme — Paper 3 Hard

1. Use the question-specific mark scheme together with the mark bands. Award up to the maximum marks as indicated.
2. Own-figure-rule (OFR) applies. If a candidate makes an error in calculation but then uses the incorrect figure appropriately and accurately in later question parts, the candidate may be fully rewarded.
3. Candidates may be penalised once for rounding errors, missing units or failure to give answers to 2 d.p.

Question 1 — Vietnamese coffee farming (perfect competition)

1. (a) (i) Outline two assumptions underlying perfect competition. [2]

Any **two** of (award [1] each):

- There is a large number of firms producing identical (homogeneous) products.
- All firms are price takers.
- There are no barriers to entry or exit.
- Producers and consumers have perfect information.
- Factors of production are perfectly mobile.

- (ii) Identify Firm X's short-run profit-maximising output. [1]

At $P = MC = \$4.00$, profit-maximising output is **50,000 kg per month**.

- (iii) Calculate Firm X's short-run economic profit at 50,000 kg. [2]

Profit = $(P - AC) \times Q = (\$4.00 - \$2.90) \times 50,000$
 = $\$1.10 \times 50,000 = \text{\$55,000 per month}$.

Valid working [1]. Correct answer with units [1]. OFR applies.

- (iv) Identify the long-run equilibrium price and output for Firm X. [2]

In LR equilibrium, firms make only normal profit — $P = \text{minimum AC}$.

From the data, minimum AC = \$2.90 at $Q = 50,000 \text{ kg}$.

LR equilibrium: $P = \$2.90/\text{kg}$, $Q = 50,000 \text{ kg/month}$.

[1] for the price; [1] for the output.

- (v) Explain, using a diagram, how Firm X moves from SR to LR equilibrium. [4]

Level		Marks
0	<i>The work does not meet a standard described by the descriptors below.</i>	0

1	<i>There is a correct diagram OR an accurate written response.</i> For a correctly labelled perfect competition diagram showing supernormal profit in SR and new firms entering, shifting market supply to the right OR an explanation that abnormal profits attract new firms [1], increasing market supply and lowering price until only normal profits remain [1].	1-2
2	<i>There is a correct diagram AND an accurate written response.</i> For a correctly labelled diagram showing the market supply shift ($S \rightarrow S'$) that brings price from \$4.00 down to \$2.90 (min AC) AND an explanation that no-barriers-to-entry leads new firms to enter until market supply has risen enough to eliminate abnormal profit [1], leaving Firm X earning only normal profit where $P = MC = \text{min AC}$ in the long run [1].	3-4

(vi) Calculate the total external cost generated in 2023. [2]

Total external cost = external cost/kg $\times$ total output
= $\$0.80 \times 1.8 \text{ billion} = \mathbf{\$1.44 \text{ billion}}$ (or \$1,440 million).

Valid working [1]. Correct answer with units [1].

(vii) Define **productive efficiency**. [2]

Level		Marks
0	<i>The work does not meet a standard described by the descriptors below.</i>	0
1	<i>Vague definition.</i> The idea that productive efficiency is about producing at low cost.	1
2	<i>Accurate definition.</i> An understanding that productive efficiency occurs when a firm produces at the minimum point of its average cost curve, so that output is produced at the lowest possible cost per unit using the fewest resources.	2

(viii) Calculate the PED and interpret. [2]

$\text{PED} = \% \Delta Q_d / \% \Delta P = -4\% / +10\% = \mathbf{-0.40}$
 $|\text{PED}| < 1$, therefore demand is **price inelastic**.

Correct calculation [1]. Correct interpretation [1].

(ix) Distinguish between allocative and productive efficiency. [2]

Marks	Level descriptor
0	<i>Does not meet standard.</i>
1	Distinction expressed vaguely, or one term explained correctly only.
2	Allocative efficiency occurs where $P = MC$ (resources are allocated so $\text{MSB} = \text{MSC}$), maximising welfare. Productive efficiency occurs where firms produce at minimum AC, using the fewest resources to produce a given output. Both are achieved in long-run perfect competition equilibrium.

(b) Recommend a policy to address externalities while supporting smallholder incomes. [10]

Possible policies include:

- Pigouvian tax on coffee production (set equal to external cost)
- Subsidies for sustainable/shade-grown coffee certification
- Direct regulation (limits on deforestation, mandatory soil-management plans)
- Guaranteed minimum prices / price floors to support smallholders
- Investment in agricultural extension services and technical training
- International cooperation on fair-trade certification

Assessment Criteria

Marks	Level descriptor
0	<i>Does not meet standard.</i>
1–2	Little understanding; irrelevant theory; no text/data.
3–5	Some understanding; theory described; superficial synthesis; limited text/data.
6–8	Appropriate theory explained; diagram(s) used; balanced evaluation; text/data applied.
9–10	Fully-developed, balanced, reasoned policy recommendation using theory, diagram and text/data to support argument.

Sample response (abridged): A Pigouvian tax of \$0.80/kg set equal to the marginal external cost would internalise the externality, raising MPC to coincide with MSC and reducing output to the socially optimal level. Revenue of up to \$1.44 bn could be hypothecated to soil-rehabilitation programmes. However, with PED of -0.40 , the tax would fall heavily on 600,000 smallholders who are price takers; their incomes would fall disproportionately. A more balanced approach combines a modest tax with a certification-linked subsidy for sustainable production, protecting smallholders while steering the industry toward sustainability.

Question 2 — Turkey balance of payments

2. (a) (i) Define current account deficit. [2]

Level		Marks
0	<i>The work does not meet a standard described by the descriptors below.</i>	0
1	<i>Vague definition.</i> The idea that a current account deficit means a country imports more than it exports.	1
2	<i>Accurate definition.</i> An understanding that a current account deficit occurs when the sum of the balance of trade in goods and services, net primary income and net secondary income is negative, i.e. debits exceed credits on the current account.	2

(ii) Calculate balance of trade in goods. [2]

Balance of trade in goods = $X - M = 256.0 - 362.0 = -\106.0 billion.

Valid working [1]. Correct answer with units and sign [1].

(iii) Calculate current account balance. [2]

CA = trade in goods + trade in services + net primary income + net secondary income
 = $(256.0 - 362.0) + (105.0 - 46.0) + (-14.0) + (+5.0)$
 = $-106.0 + 59.0 - 14.0 + 5.0$
 = **-\$56.0 billion** (current account deficit).

Valid working [1]. Correct answer with units and sign [1]. OFR applies.

(iv) State one component of the financial account. [1]

Any **one**: Foreign direct investment (FDI); portfolio investment; official reserves / reserve assets; other investment (e.g. loans and deposits).

(v) Explain two consequences of a persistent current account deficit. [4]

Marks	Level descriptor
0	<i>Does not meet standard.</i>
1–2	Two consequences stated but not explained, OR one consequence explained clearly.
3–4	Two consequences explained clearly. Possible points: depreciation of the currency; lost output and jobs in the tradeable sector; foreign debt accumulation; interest burden; loss of investor confidence; pressure on foreign reserves; possible IMF intervention; potential long-run erosion of productive capacity.

(vi) Calculate % change in the value of the lira. [2]

Jan 2023: 1 USD = 18.80 TRY → 1 TRY = $1/18.80 = \$0.0532$

Dec 2023: 1 USD = 29.50 TRY → 1 TRY = $1/29.50 = \$0.0339$

% change = $((0.0339 - 0.0532) / 0.0532) \times 100 = -36.28\%$ (depreciation).

(Accept equivalently: the lira lost about 36% of its value vs the dollar.)

Valid working [1]. Correct answer to 2 d.p. [1].

(vii) Outline the meaning of depreciation. [2]

Level		Marks
0	<i>The work does not meet a standard described by the descriptors below.</i>	0
1	<i>Vague definition.</i> The idea that depreciation is a fall in the value of a currency.	1
2	<i>Accurate definition.</i> An understanding that depreciation is a fall in the value of a currency relative to another in a floating (or managed) exchange-rate system, caused by changes in supply and demand in the foreign-exchange market.	2

(viii) Explain, using a diagram, how a depreciation may affect the current account in the long run. [4]

Level		Marks
0	<i>The work does not meet a standard described by the descriptors below.</i>	0
1	<i>There is a correct diagram OR an accurate written response.</i> For an exchange-rate diagram showing leftward shift in demand or rightward shift in supply of currency, leading to a fall in price OR an explanation that depreciation makes exports cheaper abroad and imports dearer [1], raising X and reducing M so improving the current account [1].	1-2
2	<i>There is a correct diagram AND an accurate written response.</i> For a correctly labelled exchange-rate diagram showing the depreciation AND an explanation that exports become more price-competitive while imports become dearer [1]; given sufficiently elastic demand, export revenue rises and import expenditure falls, improving the current account (subject to the Marshall-Lerner condition and the J-curve effect) [1]. Candidates who incorrectly label diagrams may receive a maximum of [3].	3-4

(ix) State whether the Marshall-Lerner condition is satisfied. [2]

$$|PED_x| + |PED_m| = 0.8 + 0.5 = 1.3 > 1.$$

Therefore the **Marshall-Lerner condition is satisfied**; depreciation should improve the current account in the long run.

Correct calculation / comparison [1]. Correct conclusion [1].

(b) Recommend a policy to reduce the current account deficit. [10]

Possible policies include:

- Contractionary fiscal policy (reduce AD and imports)
- Contractionary monetary policy (raise interest rates)
- Depreciation / managed devaluation (expenditure-switching)
- Supply-side policies to improve competitiveness (R&D; training, infrastructure)
- Trade protection (tariffs / quotas) — with caveats on retaliation
- Export promotion / trade agreements

Assessment Criteria

Marks	Level descriptor
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0	<i>Does not meet standard.</i>
1–2	Little understanding; irrelevant theory; no text/data.
3–5	Some understanding; theory described; superficial synthesis; limited text/data.
6–8	Appropriate theory explained; diagram(s) used; balanced evaluation; text/data applied.
9–10	Fully-developed, balanced, reasoned policy recommendation using theory, diagram and text/data.

Sample response (abridged): *Supply-side policies — investment in human capital and infrastructure — are the most sustainable way to reduce Turkey's CAD of \$56 bn. By raising productivity, unit labour costs fall, improving price and non-price competitiveness of Turkish exports. Since the Marshall-Lerner condition holds ($|PED_x| + |PED_m| = 1.3$), a modest further depreciation could be complementary. However, Turkey's existing 36% depreciation has already raised import inflation sharply, and supply-side gains take time to materialise. Short-term contractionary fiscal policy may therefore be needed to contain AD and imports while structural reforms take effect. Overall a combined approach — supply-side reforms with complementary macroeconomic tightening — is recommended.*